

SQL Validation Plan

Official Website and Customer Enquiry Management System for a UK Auto Parts SME

Field	Value
Client Name	Confidential Auto Parts SME
Project Phase	Phase 2 Future Enhancement - SQL and Power BI Reporting
Sprint	Sprint 4 Future SQL and Power BI Enhancement
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Document Owner	Business Analyst
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Status	Draft for Business and Data Review

Confidentiality note: This document is anonymised for BA portfolio use. It does not include any real client name, real website URL, real logo, real address, real phone number, real email, or identifiable business detail.

1. Purpose

This SQL Validation Plan defines how Phase 2 enquiry and lead tracking data should be checked before it is used for Power BI reporting. SQL means Structured Query Language. The plan focuses on business validation rules, data quality checks, and reporting readiness for the future dashboard.

2. Scope

In Scope	Out of Scope
Validation of enquiry, quote request, product category, lead status, follow-up, enquiry source, and newsletter consent data.	Building the actual SQL database, writing production stored procedures, creating a live Power BI dashboard, or connecting to real client systems.
Business-level validation rules for missing details, duplicate leads, lead status accuracy, follow-up completeness, and dashboard numbers.	Legal advice, final data protection approval, or production data migration.
Sample SQL logic for BA understanding and review with data/technical team.	Use of real customer personal data in portfolio evidence.

3. Validation Objectives

- Confirm that mandatory customer enquiry data is present before reporting.
- Identify duplicate leads so the business does not follow up the same enquiry multiple times.
- Check that lead status values are valid and consistent.
- Confirm quote request records are linked to product categories.
- Check that pending follow-ups have due dates and owners where required.
- Confirm dashboard numbers can be traced back to SQL source data.
- Check newsletter consent and unsubscribe status are captured clearly.

4. Proposed Data Areas to Validate

Data Area	Example Fields	Validation Need
Customer	Customer ID, full name, email, phone	Check mandatory contact fields and avoid duplicate customer records.
Enquiry	Enquiry ID, customer ID, enquiry date, enquiry type, source	Check every enquiry has a valid customer, date, source, and status.
Quote Request	Quote ID, enquiry ID, category ID, vehicle make/model/year, part details	Check quote requests are linked to enquiry and product category.
Product Category	Category ID, category name, active flag	Check only approved categories are used in reporting.

Data Area	Example Fields	Validation Need
Lead Status	Lead ID, enquiry ID, status, status date	Check status values such as New, Contacted, Quote Sent, Won, Lost, Closed.
Follow-up Note	Follow-up ID, enquiry ID, owner, due date, note	Check pending follow-ups have owner and due date.
Enquiry Source	Source ID, source name	Check source names are consistent for reporting.
Newsletter Subscription	Email, consent date, unsubscribe flag	Check consent and unsubscribe status are available.

5. SQL Validation Scenario List

ID	Validation Area	Business Rule	Expected Result	Priority
SQL-VAL-01	Missing contact details	Customer enquiry must include customer name and email. Phone is optional.	Records with blank name or email should be flagged for correction.	High
SQL-VAL-02	Invalid email format	Email address should contain a valid format before it is used for follow-up.	Invalid email records should be listed for review.	High
SQL-VAL-03	Duplicate leads	Same email plus same product category within short period may be duplicate.	Possible duplicate leads should be flagged, not automatically deleted.	Medium
SQL-VAL-04	Missing product category	Quote requests should have product category selected.	Quote records without category should be flagged.	High
SQL-VAL-05	Lead status accuracy	Lead status should use approved status list only.	Any unknown status should be flagged.	High
SQL-VAL-06	Pending follow-up completeness	Pending follow-up should have owner and due date.	Records missing owner/due date should appear in exception list.	High
SQL-VAL-07	Closed lead follow-up check	Closed leads should not appear as pending follow-up.	Closed leads appearing as pending should be flagged.	Medium
SQL-VAL-08	Newsletter consent	Newsletter subscription should have consent date and unsubscribe status.	Missing consent date should be flagged.	High
SQL-VAL-09	Dashboard total enquiries	Total enquiry count in Power BI should match SQL enquiry count for selected date range.	Dashboard and SQL numbers should match.	High
SQL-VAL-10	Monthly trend readiness	Enquiry date should be available for monthly trend reporting.	Records without enquiry date should be flagged.	Medium

6. Sample SQL Logic for BA Review

These are sample validation checks for discussion with the technical team. They are not production-ready scripts.

```
-- Check missing mandatory contact details
SELECT enquiry_id, customer_name, email_address
FROM enquiry_view
WHERE customer_name IS NULL
      OR email_address IS NULL;
```

```
-- Check possible duplicate leads
SELECT email_address, product_category, COUNT(*) AS enquiry_count
FROM enquiry_view
GROUP BY email_address, product_category
HAVING COUNT(*) > 1;
```

```
-- Check invalid lead status values
SELECT enquiry_id, lead_status
FROM lead_status_view
WHERE lead_status NOT IN ('New', 'Contacted', 'Quote Sent', 'Won', 'Lost', 'Closed');
```

```
-- Check pending follow-ups without owner or due date
```

```

SELECT enquiry_id, follow_up_owner, follow_up_due_date
FROM follow_up_view
WHERE lead_status = 'Pending Follow-up'
AND (follow_up_owner IS NULL OR follow_up_due_date IS NULL);

```

7. Dashboard Number Reconciliation Checks

Power BI Metric	SQL Source Check	Expected Validation Outcome
Total Enquiries	Count enquiry_id from enquiry table for selected date range.	Power BI total should match SQL total.
Quote Requests	Count quote_request_id from quote request table.	Power BI quote count should match SQL count.
Top Product Categories	Group quote requests by product category.	Top categories should match SQL grouped result.
Pending Follow-ups	Count enquiries with pending follow-up status and due date.	Dashboard pending number should match SQL exception/lead view.
Conversion Rate	Won leads divided by total qualified leads.	Formula should be agreed and consistent in SQL and Power BI.
Monthly Enquiry Trend	Group enquiries by enquiry month.	Trend chart should match SQL monthly count.

8. Roles and Responsibilities

Role	Responsibility
Business Analyst	Defines validation scenarios, business rules, expected results, and supports review of exceptions.
Product Owner	Confirms priority of reporting checks and acceptance criteria.
Business Owner	Confirms data meaning, lead status definitions, and follow-up rules.
Data/SQL Developer	Creates technical SQL queries, views, and validation scripts.
Power BI Developer	Uses validated SQL data for dashboard visuals and reconciles dashboard numbers.
Compliance Owner/Advisor	Confirms data usage, consent, retention, access, and reporting rules where needed.

9. Acceptance Criteria

- Mandatory data validation checks are documented for enquiry and lead tracking data.
- Duplicate lead identification approach is documented as possible duplicate, not automatic deletion.
- Lead status approved values are documented and ready for technical validation.
- Follow-up validation checks are documented for owner, due date, and pending status.
- Newsletter consent and unsubscribe data checks are included.
- Dashboard reconciliation checks are documented for key Power BI metrics.
- Open compliance points are clearly marked for business/legal confirmation.

10. Assumptions and Open Questions

Type	Item
Assumption	Phase 2 uses sample/anonymised data for portfolio evidence.
Assumption	Final SQL table design will be owned by the technical team.
Open Question	What exact lead statuses should the business use in the final process?
Open Question	How long should enquiry and follow-up data be retained?
Open Question	Who can access customer contact details in the dashboard?
Open Question	Should Power BI show customer-level details or only aggregated numbers?

11. Portfolio Note

This document is suitable for portfolio evidence as a BA artefact. It shows future SQL validation planning and Power BI reporting readiness. It should be presented as requirements and validation planning work, not as a delivered production database or live dashboard.